

# Anas Houssaini

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## EDUCATION

### McGill University

Montreal, CA

MSc - Thesis - Computer Science @ Intelligent Machines Lab under Dr. Meger & Dr. Lin, GPA: 4.0 Aug 2024 – Present

### Concordia University

Montreal, CA

Honors Computer Science, Minor in Philosophy - Dean's List — Distinction

Sep 2020 – April 2024

### Cambridge College of Arts Science

Cambridge, UK

Certificate Physics & Economics, GPA: 4.0 - Honors Award in Economics - Rank: 1/63

Sep 2019 – May 2020

## EXPERIENCE

### Amazon FAR | Applied Scientist II Intern, Montreal, Canada

May 2026 – Present

- Main author on a functional bimanual retargeting project.

### Kinova Robotics | AI & Robotics Intern, Montreal, Canada

May 2025 – October 2025

- Trained and deployed state-of-the-art vision-language-action models on the Link 6 industrial arm for complex picking and manipulation tasks.
- Sole-authored the ROS 2 stack for the Link 6 arm, delivering a client-ready package shipped to partners including Locus Robotics and research labs.
- Built an end-to-end data collection pipeline using HTC Vive trackers and converted teleoperated demonstrations into **LeRobot** and **RoboMimic** formats for VLA and behavior cloning training.
- Built data collection and deployment workflow for a \$1M government-funded greenhouse picking project.

### McGill University | Summer Research Assistant, Montreal, Canada

May 2024 – August 2024

- [Rebuilt the \\$200k Movo mobile manipulator](#) to support large-scale ML workloads, modern robotics hardware, and a rebuilt ROS architecture for navigation, manipulation, and low-level control.
- [Created a Movo simulation environment in NVIDIA Isaac Sim](#) with manipulation tasks and a training-to-deployment pipeline.

## CURRENT RESEARCH PROJECTS

### Hand Object Interaction Retargeting

On-going

- On-going internship project at Amazon FAR (Main Author).

### Contact Aware Interactive World Models

On-going

- On-going project using FlexiTac sensor in IWS (Co-Author)
- Project Supervised by Professor **Yunzhu Li** @ **Columbia University** & **Hsiu Chin Lin**

## RESEARCH PUBLICATIONS

### [DriftQL: Drift Q-Learning for Offline Reinforcement Learning](#) (NeurIPS 2026)

2026

- **Main author** and work supervised by Scott **Fujimoto**, Hsiu-Chin **Lin**, and David **Meger**.
- Proposed a new offline RL paradigm inspired by drift models, using behavioral regularization with critic-driven policy improvement.
- Developed a one-step policy method that samples actor actions, compares them against dataset actions and nearby actor samples, and trains the actor toward corrected drifted targets.
- Achieved state-of-the-art results on OGBench and D4RL with robustness to noisy offline datasets, requiring no denoising chains, ODE solvers, auxiliary networks, or distillation.

### [DexSuite: Unified Simulation Framework for Dexterous Manipulation](#) (Submitted to RA-L)

Release July 2026

- **Main author and project lead** for DexSuite, a unified dexterous robot simulation framework for single-arm and bimanual manipulation with multi-finger hands and mixed-material physics, supervised by David **Meger**, Hsiu-Chin **Lin**, Gregory **Dudek**, and Doina **Precup**.
- Released a teleoperation and data toolchain, a 2,000-demo Manus glove expert dataset, and a curated benchmark with **21 environments** and **20+** robot-gripper configurations for evaluating dexterous policy learning.

### Toward Hardware-Agnostic Quadrupedal World Model via Morphology Conditioning (CORL 2026)

2026

- Co-authored work on morphology-conditioned quadrupedal WM's for hardware-agnostic learning and transfer, supervised by Glen **Berseth**, Marco **Hutter**, and Hsiu-Chin **Lin**.
- Contributed to training and deploying quadrupedal policies across different robot morphologies.

#### **Tactile Modality Fusion for VLA Models ( ECCV 2026)**

2026

- Co-authored work on integrating tactile sensing into VLA's for robotic manipulation.
- Worked on the preliminary architecture for tactile-modality fusion in VLA models.

#### **Contractive Diffusion Policies (ICLR 2026)**

Jun – Sep 2025

- Contributed mathematical derivations and validation of contraction conditions, helping shape a stable training objective for policy diffusion.
- Co-developed the real-world deployment and data-collection platform for robot experiments, including on-robot execution and synchronized sensor logging.

#### **VOCALoco: Viability-Optimized Cost-Aware Adaptive Locomotion (ICRA 2026)**

Jan – Apr 2025

- Helped build the end-to-end deployment pipeline to run terrain-specialized locomotion policies on ANYmal-D, integrating heightfield perception with viability and cost-of-transport modules for safe, energy-aware stair ascent and descent.

### RESEARCH PROJECTS

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#### **MOVO Rebuild: Resurrection of a \$200k Mobile Manipulator**

May 2024 – Aug 2024

- Led the complete hardware/software resurrection of a \$200k MOVO mobile manipulator, a 10-year-old undocumented prototype with dual 7-DOF Kinova arms.
- Reverse-engineered the entire system, replacing failed proprietary embedded boards/batteries and implementing a new ROS 2 omni-drive controller by decoding the motor CAN bus protocol.
- Modified the Kinova C++ SDK to enable unsupported dual-arm Ethernet control and integrated a Jetson Orin and Realsense cameras, transforming the dead hardware into an AI research platform.
- Built a high-fidelity digital twin of the dual-arm MOVO in **Isaac Lab**, designing custom bimanual manipulation environments and a sim-to-real training pipeline.

#### **NanoStride: Founded AI/Robotics Undergraduate Research Group (\$25k funded)**

Feb 2022 – May 2024

- Founded the first AI/Robotics group at Concordia, growing membership to 50+ students.
- Designed and developed a 5'8ft robot that serves 1500+ students monthly as a front desk at Concordia University.
- Developed and trained ML models for the robot's conversational and vision abilities, and human-robot interaction.

#### **Propulsion and Avionics Lead @ Space Concordia**

Jul 2021 – Dec 2021

- Led the avionics team in developing the end-to-end electronic system for Starsailor, the world's most powerful student-built rocket (13m, 123km apogee target).
- Designed the rocket's data acquisition (DAQ), high-speed telemetry, and power distribution systems from scratch, handling data from all critical flight sensors.
- Oversaw the integration of the flight computer, sensor suite (IMU, GPS, barometer), and actuation circuits for recovery systems (parachute deployment).

#### **Ares JS1: Custom-Built Lightweight Racing Car**

Mar 2017 – May 2019

- Designed and built a 3m x 1.7m lightweight racing car from scratch over two years, resulting in a functional vehicle with a top speed of 160 km/h.
- Engineered and fabricated the 150kg ionized steel and aluminum chassis, teaching myself welding, aerodynamics, and mechanical design principles.
- Integrated a 160hp 2.2L VTEC engine and custom-designed the entire mechanical system, including precision suspension geometry for high-speed stability.

### TECHNICAL SKILLS

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**Programming Languages:** Python, Java, C, C++, SQL, Javascript, C#, Bash

**Applications:** Kubernetes, Docker, Git, Isaac Lab, Google Cloud, Jupyter, AWS, Jira, Figma, Adobe Suite

**Frameworks:** Numpy, Pandas, Pytorch, Scikit, Keras, ROS, Mujoco, TensorFlow, React.JS, Node.Js, ASP.NET

**Interests:** Building robotic arms, drones, training models, and filmmaking **Languages:** French, English and Arabic